Laravel Framework

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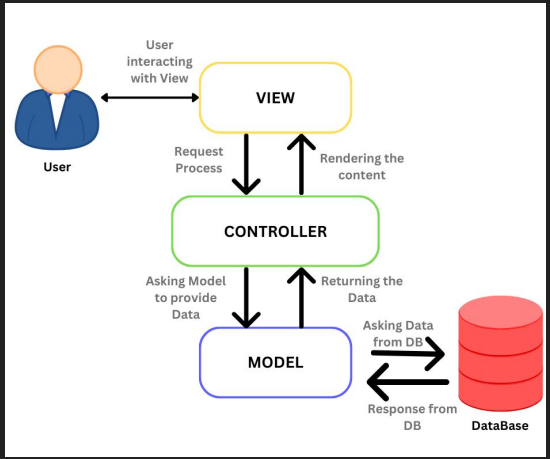
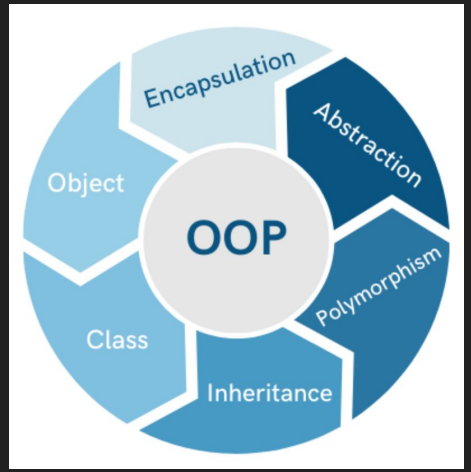
Sebelum Belajar

- Understand Basic PHP
- Understand Basic HTML & CSS
- Understand MVC Design Pattern
- Understand OOP





Agenda

- Intro
- Route
- View (Blade Template Engine)
- Dummy Model + Collections
- Migration
- Seeder + Factory + Faker
- Create Model
- Controller
- Relations
- ORM Problems (N+1)
- Searching & Pagination
- Authentication
- Create BE UI
- CRUD + Validation
- Authorizations

Apa Itu Laravel?

Laravel is a **web application framework** with expressive, elegant syntax. We've already laid the foundation, freeing you to create without sweating the small things.

~ Laravel Doc~

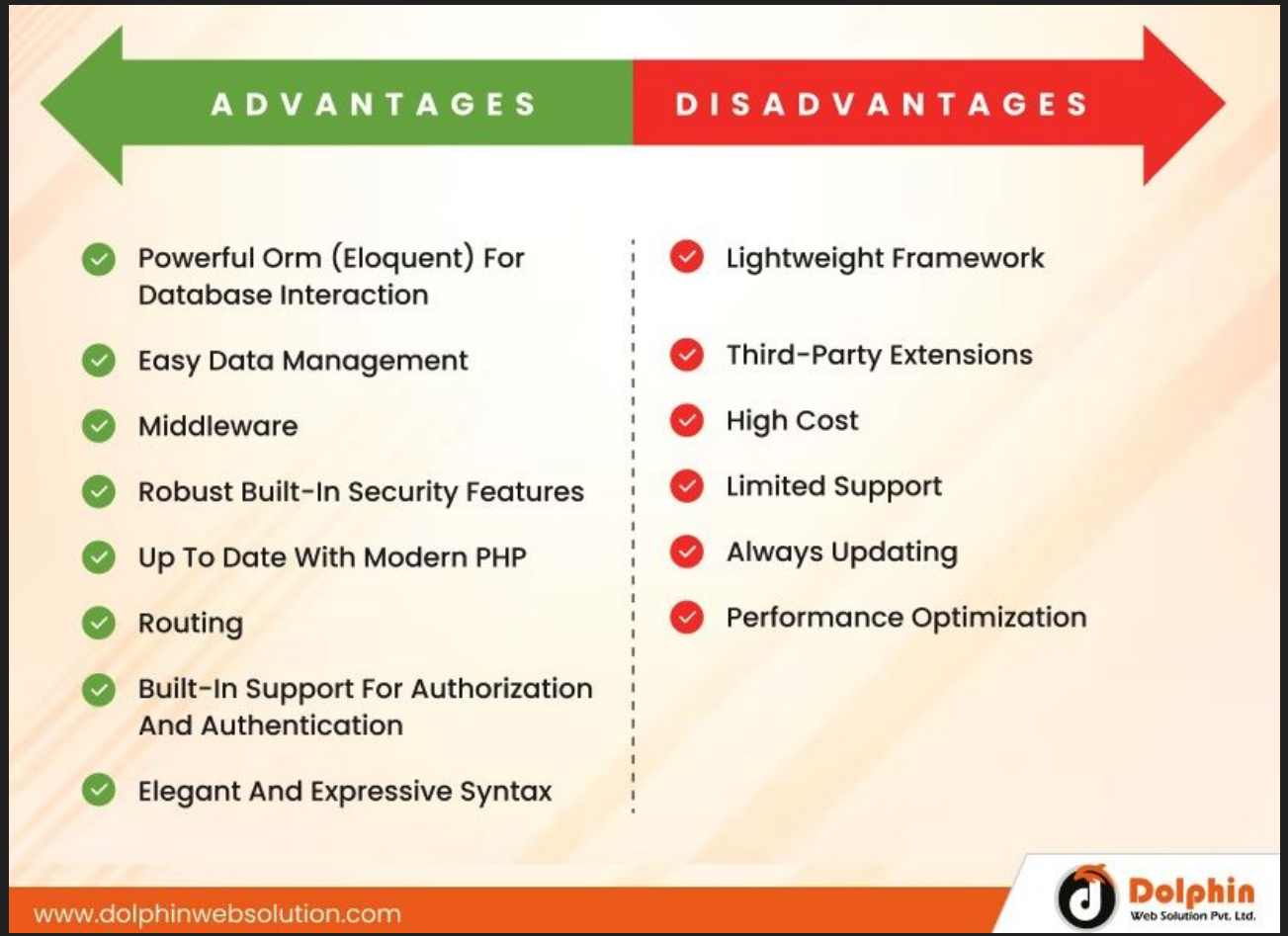
Fitur Laravel

- Authentication
- Libraries & Modular
- Artisan
- Template Engine
- Effective ORM
- MVC Architecture
- Secure Migration
- Unique Unit Testing
- Intact Security

Kelebihan & Kekurangan

Kelebihan Lain

- Good documentation
- Good community
- Library Support



Source >

PHP Framework Comparison

https://www.excellentwebworld.com/best-php-frameworks

Persiapan

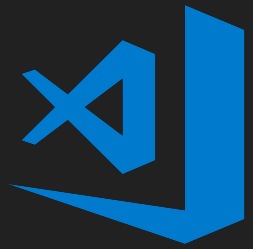
- PHP ^8.1
- DB ~ Mysql ^5.7
- Composer 2 (php dependency management)
- DBMS
- Terminal
- Editor Code (vscode / phpstorm)

VSCode

PHPStorm

- Syntax
- Auto Complete
- Git
- Debugging
- Speed Up

VsCode Extensions



VSCode

- PHP Elephense
- Laravel Artisan
- Laravel Snippet
- Laravel Blade Snippet
- Blade Space
- Laravel Goto View

Hands On

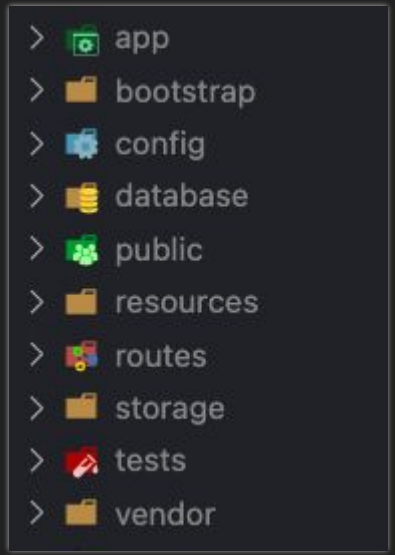
https://laravel.com

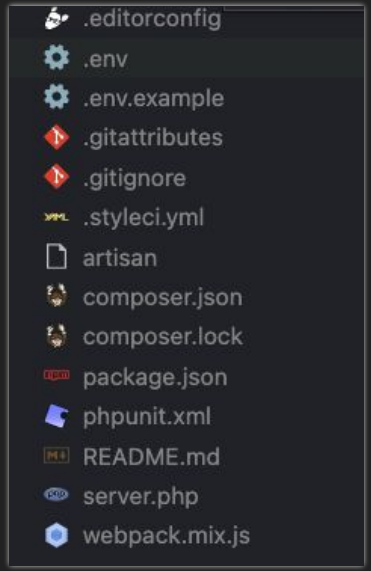
Installation & Configuration

- Install PHP & mysql (Manually, Xampp, Wamp, dll)
- Install Composer / download composer https://getcomposer.org/download
- Install Laravel Project (https://laravel.com/docs)
- Run the application

```
$ php artisan serve
```

Directory Structures





.ENV Configuration

https://laravel.com/docs/8.x/configuration#environment-configuration

Composer.Json & Lock File

Routing

address

Controller Class

Function

```
Route::get('/user/{id}', [UserController::class, 'index']);
```

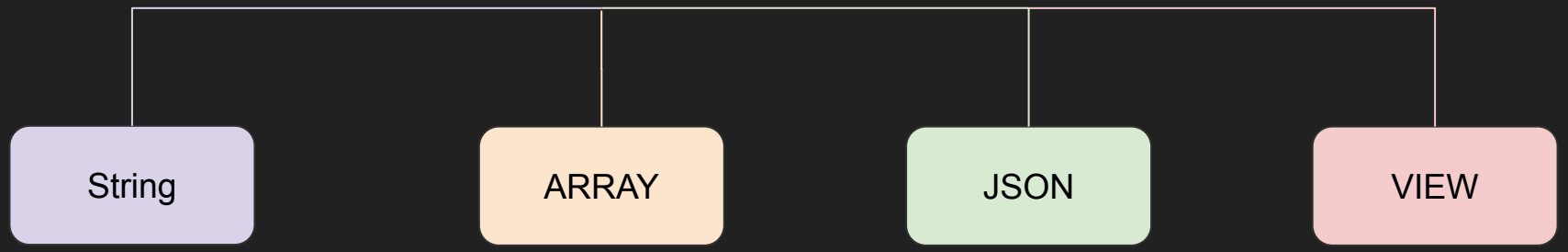
Request Method / others
get, post, put, delete, dll

Closure Function

```
//routing dasar
use Illuminate\Support\Facades\Route;
Route::get('/greeting', function () {
   return 'Hello World';
});

//dengan required parameter
Route::get('/user/{id}', function ($id) {
   return 'User '.$id;
});
```

```
use App\Http\Controllers\UserController;

// <= L7
Route::get('/user', 'UserController@index']);

//Memanggil fungsi controller dengan static class (L8)
Route::get('/user', [UserController::class, 'index']);
```

Route Response



View

User Interface yang menampilkan informasi kepada pengguna.

View dalam laravel di-handle dengan template engine blade.

Controller

Untuk menggantikan semua penanganan request dan logic dalam closures di dalam routing dan agar lebih terorganisir, maka semua logic & handler diletakkan pada kelas controller yang tersimpan dalam direktori *./app/http/controllers*.

Blade Template Engine

- Template Engine

  Tools untuk membantu dalam mengelola tampilan dalam sebuah framework

- Blade

  Template Engine dari framework laravel. Fungsi yang paling sederhana adalah membuat variable, looping, conditional, layouting, component, parsial, dll. Extension file berupa *".blade.php"*

  ```
  view_name.blade.php
  ```

  https://laravel.com/docs/8.x/blade

Middleware

Mekanisme yang disediakan untuk melakukan filter pada http request yang masuk.

Membuat middleware `php artisan make:middleware MiddlewareName`

https://laravel.com/docs/8.x/middleware#main-content

```php
public function handle($request, Closure $next)
{
    if ($request->input('token') !== 'my-secret-token') {
        return redirect('home');
    }

    return $next($request);
}
```

DB Migration & Model / Factory Model

Seperti version control untuk database. Dengan migration akan memberikan kemudahan untuk pengelolaan database dan kolaborasi dengan tim. Migration memungkinkan developer membagi setiap perubahan yang terjadi pada database dengan cara yang mudah dan menghindari masalah..

https://laravel.com/docs/8.x/migrations

Query Builder

Untuk menjalankan / mengeksekusi query di Framework laravel.

https://laravel.com/docs/8.x/queries

Eloquent

Eloquent merupakan ORM (object relational mapper) atau database abstraction layer yang memudahkan dalam berinteraksi dengan database / tabel. Dalam menggunakan Eloquent, setiap tabel terikat dengan `Model` yang digunakan untuk berinteraksi dengan tiap tabel tersebut.

ORM merupakan teknik untuk merubah table menjadi objek yang memiliki properti yang sama dengan tabelnya dan memiliki beberapa fitur untuk memudahkan dalam berinteraksi dengan database seperti, query, transaction, pooling, stream dll.

https://laravel.com/docs/8.x/eloquent

Table: Category

Id	name
1	Politik
2	Teknologi
3	Ekonomi

HasMany

BelongsTo

Table: posts

id	title	desc	categori_id
1	Xiaomi	desc	1
2	Realme	desc	2
3	Samsung	desc	3

Eloquent Relation II

Has Many

Table: phones

id	name	founded	description
1	Xiaomi	2010	Desc ...

Table: kinds

Id	phone_id	name
1	1	Redmi Note 7

Table: colors

id	kind_id	color
1	1	Black
2	1	White
3	1	Blue

Eloquent Relation III

Has One

Table: Users

Id	phone_id	name
1	1	Redmi Note 7
2	1	Redmi Note 10

Table: Profile

id	kind_id	date
1	1	2018
2	2	2021

Eloquent Relation IV

Many to Many

Table: phones

id	name	founded	description
1	Xiaomi	2010	Desc ...
2	Samsung	1938	Desc …

Table: products

id	name
1	smartphone
2	tablet
3	smartwatch

Eloquent Relation IV

Many to Many

Table: phones

id	name	founded	description
1	Xiaomi	2010	Desc ...
2	Samsung	1938	Desc …

Pivot Table: phone_products

car_id	kind_id
1	1
1	2
3	2
3	3

Table: products

id	name
1	smartphone
2	smartwatch
3	tv

Validation Request

```
$validated = $request->validate([
      'title' => 'required|unique:posts|max:255',
      'body' => 'required',
  ]);
```

Create Form Request

```
Php artisan make:requests [request_name]
```

File Upload

Laravel menyediakan fitur upload file yang sangat mudah. Fitur ini dapat terintegrasi dengan beberapa driver yaitu: local filesystem, sFTP, AWS S3, dll.

Pagination

https://laravel.com/docs/9.x/pagination#using-bootstrap

Authentication

Generate UI / view untuk kebutuhan authentication

`` `php artisan ui:auth` ``

Mailer

Membuat Class Mailer

```
php artisan make:mail [mail_name_class]
```

Mengirim email

```
Mail::to($recipient)->send(new ExampleMail($param));
```

Job / Queue

https://laravel.com/docs/8.x/queues#introduction

Localization

https://laravel.com/docs/8.x/localization#introduction

Laravel Best Practice

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